

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 11: Sampling Theory

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1 Sampling Theory: $x(t) \longleftrightarrow x[n]$

Consider the analog signal

$$x(t) \longleftrightarrow X(j\Omega).$$

Sample uniformly at an interval T :

$$x[n] = x(t)|_{t=nT}, \quad n \text{ integer}, \quad x[n] \longleftrightarrow X(e^{j\omega}).$$

Question.

What is the relationship between $X(j\Omega)$ and $X(e^{j\omega})$?

Recall:

$$\begin{aligned}
 x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega \\
 \Rightarrow x[n] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega nT} d\Omega \\
 &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \int_{-\pi/T}^{\pi/T} X\left(j\left(\Omega + \frac{2\pi k}{T}\right)\right) e^{j\left(\Omega + \frac{2\pi k}{T}\right)nT} d\Omega \\
 &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \int_{-\pi/T}^{\pi/T} X\left(j\left(\Omega + \frac{2\pi k}{T}\right)\right) e^{j\Omega nT} d\Omega.
 \end{aligned}$$

Remark

(addendum to orig lecture note).

The second line splits the real line into intervals of length $\frac{2\pi}{T}$ centered at $\frac{2\pi k}{T}$.
 The third line uses $e^{j\frac{2\pi k}{T}nT} = e^{j2\pi kn} = 1$ for integers k, n .

Change of variables:

$$\omega = \Omega T \iff \Omega = \frac{\omega}{T}, \quad \Omega = \pm \frac{\pi}{T} \Rightarrow \omega = \pm \pi, \quad d\omega = T d\Omega.$$

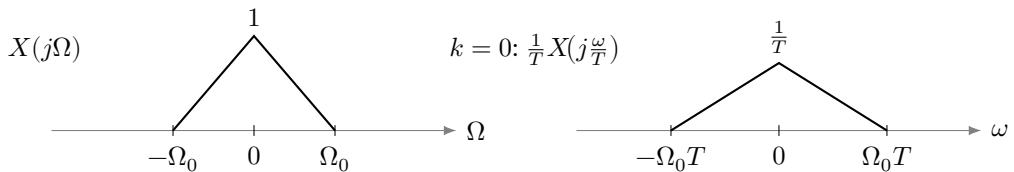
2 1.1 The Sampled Spectrum

$$\begin{aligned} x[n] &= \frac{1}{2\pi T} \sum_{k=-\infty}^{\infty} \int_{-\pi}^{\pi} X\left(j\frac{\omega}{T} + j\frac{2\pi k}{T}\right) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \underbrace{\left[\frac{1}{T} \sum_{k=-\infty}^{\infty} X\left(j\frac{\omega}{T} + j\frac{2\pi k}{T}\right) \right]}_{X(e^{j\omega})} e^{j\omega n} d\omega \end{aligned}$$

Sampled spectrum.

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j\Omega) \Big|_{\Omega = \frac{\omega}{T} + \frac{2\pi k}{T}}$$

Example: a triangular $X(j\Omega)$ of height 1 supported on $[-\Omega_0, \Omega_0]$. The $k = 0$ term is $\frac{1}{T} X(j\frac{\omega}{T})$, a triangle of height $\frac{1}{T}$ on $[-\Omega_0 T, \Omega_0 T]$:

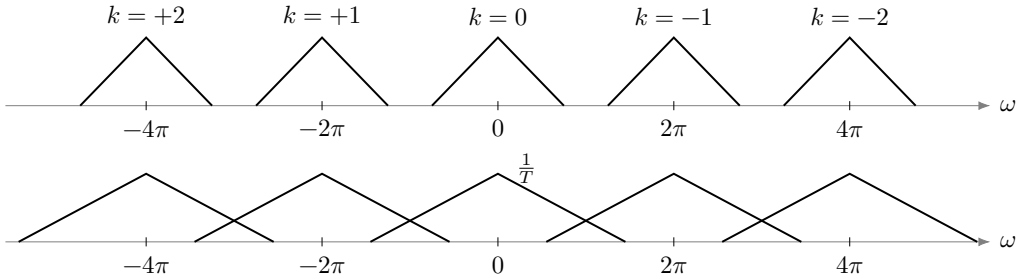


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The $k = -1$ term and the sum over k .

$$k = -1 \Rightarrow \frac{1}{T} X\left(j\frac{\omega}{T} - j\frac{2\pi}{T}\right) = \frac{1}{T} X\left(j\left(\Omega - \frac{2\pi}{T}\right)\right) \Big|_{\Omega=\frac{\omega}{T}},$$

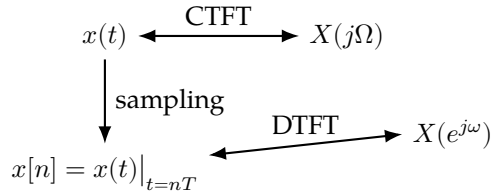
i.e., the triangle moves to $\Omega = \frac{2\pi}{T}$ (support $[\frac{2\pi}{T} - \Omega_0, \frac{2\pi}{T} + \Omega_0]$), which is $\omega = 2\pi$ (support $[2\pi - \Omega_0 T, 2\pi + \Omega_0 T]$). Each k contributes one copy centered at $\omega = -2\pi k$:



Top: copies do not overlap. Bottom: wider $X(j\Omega)$ (or larger T), so neighboring copies overlap.

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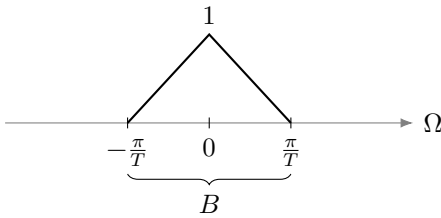
2 Summary: CTFT, Sampling, and DTFT



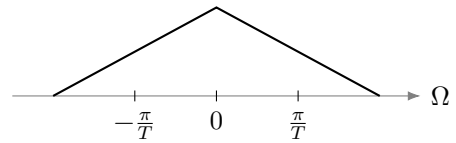
$$X(e^{j\omega}) = \underbrace{\frac{1}{T} \sum_{k=-\infty}^{\infty}}_{\text{rescaling periodization}} X(j\Omega) \Big|_{\underbrace{\Omega = \frac{\omega}{T} + \frac{2\pi k}{T}}_{\text{stretching of the axis}}}$$

Two cases of interest:

- $X(j\Omega) = 0$ whenever $|\Omega| \geq \frac{\pi}{T}$ (bandlimited signal with bandwidth $B \leq \frac{2\pi}{T}$).
- $X(j\Omega) \neq 0$ for some $\Omega \geq \frac{\pi}{T}$ or $\Omega \leq -\frac{\pi}{T}$.



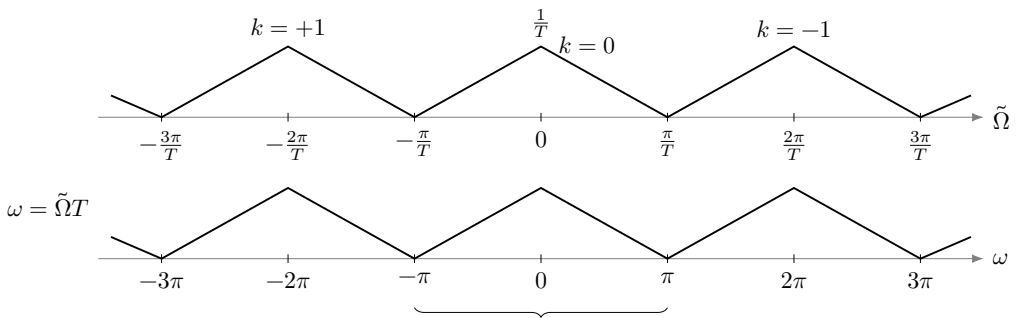
bandlimited

not bandlimited to $\frac{\pi}{T}$

5 2.1 The Bandlimited Case

Consider

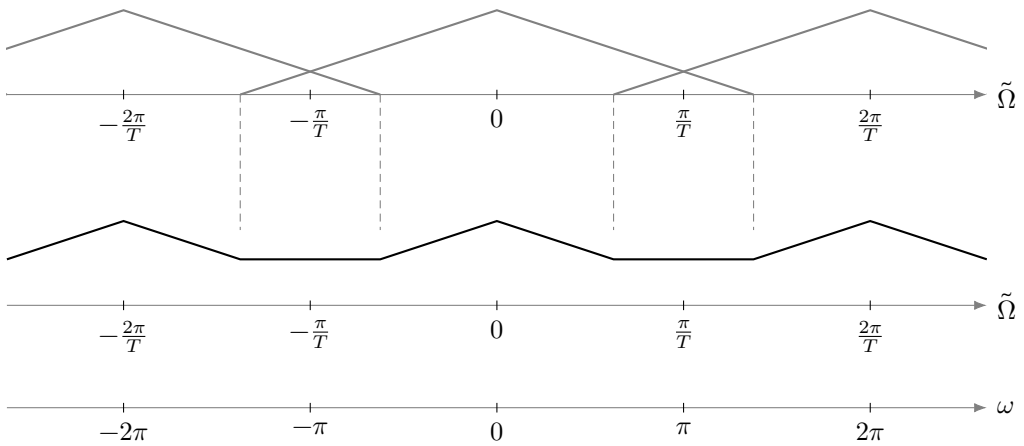
$$\frac{1}{T} \sum_{k=-\infty}^{\infty} X(j\Omega) \Big|_{\Omega=\tilde{\Omega}+\frac{2\pi k}{T}}.$$



On $-\pi \leq \omega \leq \pi$, the **shape of the original CTFT $X(j\Omega)$ is preserved** (and $x(t)$ can be perfectly recovered from $x[n]$ if we follow appropriate steps).

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2.2 The Other Case (Aliasing) and the Sampling Theorem



Overlapping copies add up, so the sum no longer looks like one copy of $X(j\Omega)$.

The **CTFT shape is not preserved** (EE 592).

Sampling theorem.

A signal can be recovered from its samples provided that the signal satisfies

$$X(j\Omega) = 0 \quad \text{for } \forall |\Omega| \geq \frac{\pi}{T},$$

where T is the sampling interval.

- Nyquist – Shannon – Whittaker – Kotelnikov

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3 Sampling Rate and the Nyquist Criterion

Converting from Ω (radians/sec) to f (Hz = 1/sec):

$$f = \frac{\Omega}{2\pi \text{ radians}}.$$

Largest signal frequency:

$$|f_{\max}| < \frac{(\pi/T)}{2\pi} = \frac{1}{2T}.$$

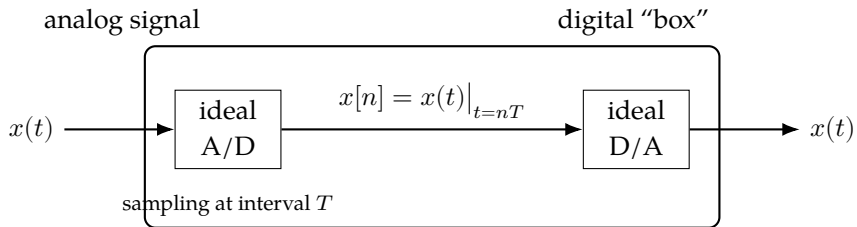
Let $f_s \triangleq \frac{1}{T}$ be the **sampling rate** (sampling frequency).

Nyquist criterion (Nyquist condition).

For no aliasing,

$$f_s > 2|f_{\max}|.$$

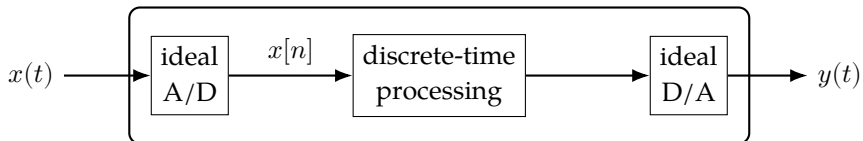
- Frequency $2|f_{\max}|$: **Nyquist rate**.
- Frequency $f_s/2$: **Nyquist frequency**.



This works if $x(t)$ is appropriately bandlimited.

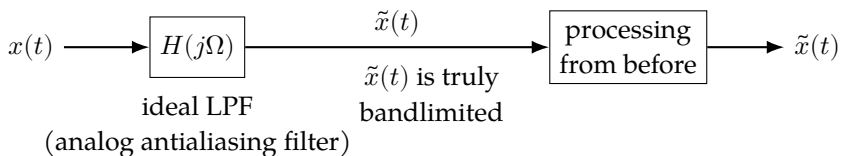
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3.1 Discrete-Time Processing of Analog Signals



Equivalent to analog processing (but is much easier).

What if $x(t)$ is not bandlimited?



Example.

$$g_1(t) = \cos(6\pi t) \quad \longleftrightarrow \quad G_1(j\Omega) = \pi[\delta(\Omega - 6\pi) + \delta(\Omega + 6\pi)],$$

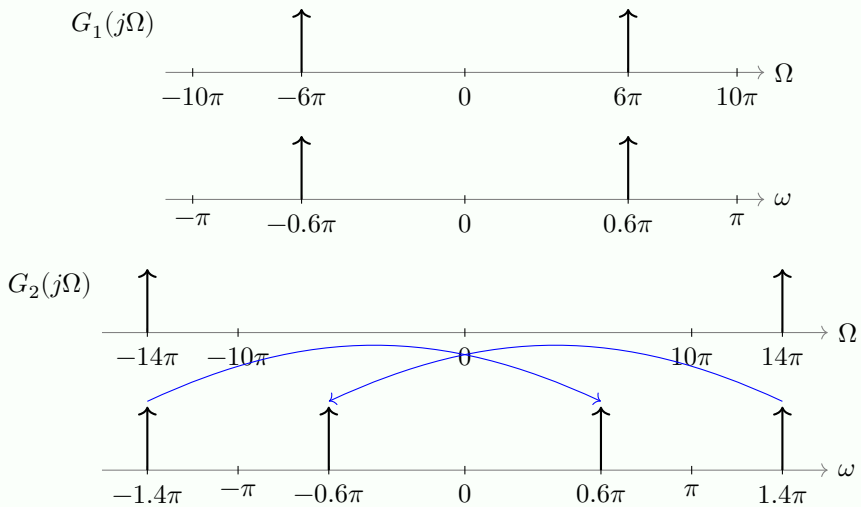
$$g_2(t) = \cos(14\pi t) \quad \longleftrightarrow \quad G_2(j\Omega) = \pi[\delta(\Omega - 14\pi) + \delta(\Omega + 14\pi)],$$

$$g_3(t) = \cos(26\pi t) \quad \longleftrightarrow \quad G_3(j\Omega) = \pi[\delta(\Omega - 26\pi) + \delta(\Omega + 26\pi)].$$

These are 3 Hz, 7 Hz, and 13 Hz signals. Sample at interval $T = 0.1$ sec:

$$\text{sampling freq} = \frac{1}{T} = 10 \text{ Hz} = 20\pi \text{ rad/sec.}$$

Signal content above 5 Hz will be aliased.



For $G_3(j\Omega)$: $\Omega = \pm 26\pi \Rightarrow \omega = \pm 2.6\pi \Rightarrow \pm 0.6\pi$.

We can't tell the difference between any of these signals! $g_1[n] = g_2[n] = g_3[n]$.

Remark*(addendum to orig lecture note).*

Directly in time: $g_2[n] = \cos(1.4\pi n) = \cos(1.4\pi n - 2\pi n) = \cos(-0.6\pi n) = \cos(0.6\pi n) = g_1[n]$, and likewise $g_3[n] = \cos(2.6\pi n) = \cos(2.6\pi n - 2\pi n) = \cos(0.6\pi n)$. Only g_1 (3 Hz < 5 Hz) satisfies the Nyquist criterion, so 3 Hz is the frequency an ideal D/A would reconstruct for all three.

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Secs. 2.5, 3.8, 3.9
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 3